

RFID Technology: A Complete Overview

RFID technology allows non-contact transfer of information (much like the familiar barcode), making it effective in manufacturing and other hostile environments where barcode labels could not survive. Take a look at how it works and what are its pros and cons

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Radio-frequency identification (RFID) technology involves the use of electromagnetic or electrostatic coupling in the RF portion of the electromagnetic spectrum to uniquely identify an object, animal or

person. It has established itself in a wide range of markets including livestock identification and automated vehicle identification because of its ability to track moving objects. The technology has also become a primary component of automated data collection, identification and analysis systems worldwide.

Architecture and working

An RFID system consists of three components: transceiver (often combined into the reader), some sort of data processing equipment, such as a computer, and a transponder (the tag). A typical RFID system is shown in Fig. 1.

RFID tag, RFID tag, usually known

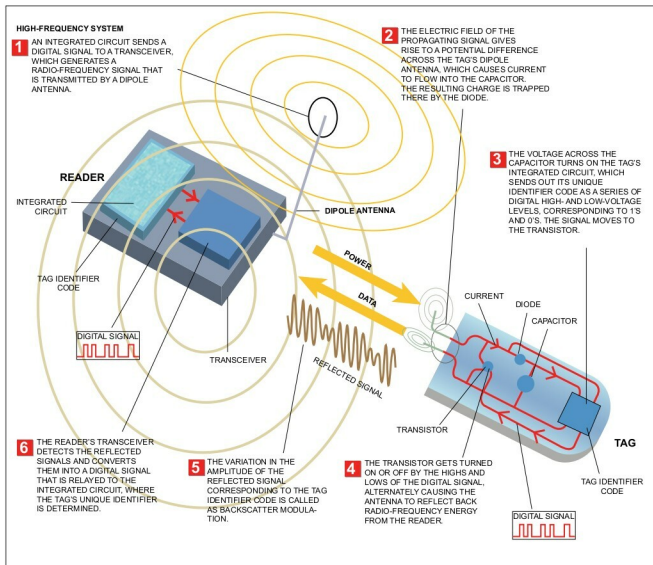


Fig. 1: RFID architecture and working